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H-SCOOP

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BITS & PIECES

A REMARKABLE EXPERIENCE

I spent a full day at the Heath Company, Monday February 22nd. Heath Company arranged to have myself and Charlie Floto, editor of BUSS visit them to see the plant operations and see their new products. Before I go any further, please, don't anyone call or write to find out what I saw. We had to sign an agreement stating we will not disclose what we saw. I will say that there's some mighty exciting things happening at Heath and some excellent and unexpected products coming from them in the near future. I will also say that anyone who is too anxious to wait for the new computer and chooses an IBM or TRASH instead, will regret it evermore! The wait will not be too much longer, and well worth it. You'll also find an unbelievable amount of compatibility with their new machine.

Why am I telling you this? Because I feel with support and concern like this from Heath/Zenith, we have nothing to loose. Their main reason of letting us see this stuff was to stop untrue rumors which may cause more harm than good. You show me another company like that. Or show me another which releases source code for their operating system! Or another that has the support that Heath does. We definitely have picked a winner with Heath. They are growing very fast, and it's nice to be taking the ride with them.

As to current developments, Barry Watzman showed me a new board for the H8 to be released later this year. It will be a DD disk controller for 5" drives and also support hard disks for the H8. Barry did indicate that this was currently the last board Heath was planning for the H8. He also indicated that the new serial card for the H89 (with the IBM compatibility and the parallel port) was currently the last planned board for the H89 from Heath. Bill Johnson, president of Heath, assured me that support for both machines would continue. He said parts will be handled for 7 years as required by law after manufacturing and sales of the machine ceases, and technical support will continue on both machines.

One of the major complaints against Heath as reported in the returned questionnaires was the hard time to get technical support because phone lines are "always busy". Bill Johnson told me that the lines are monitored, and if enough busy signals are reported, more technical help will be hired. He did indicate that it must be an average trend. As the computer manuals say, call in the middle of the week for less traffic. If the lines are busy on Monday and Fridays, but quite open the rest of the time, to hire more help would just be wasteful as they would be sitting around most of the time. I tend to agree.

SURVEY RESULTS

I have compiled the returned questionnaire results. I had a 23% response to my questionnaires, which is not bad. The normal response is supposed to be 5%, for normal people. You are not normal people, and I expected a 100% response. If it's too much work to take 10 minutes to answer the questionnaire, and spend 20 cents to mail it; then don't complain when you don't get what you want from this end. It's only as I receive your feedback that I can give you what you want, to make you happy!

Anyway, as I said last year when the questionnaires were returned, you can't please everybody all the time. Again, I was torn between extremes. The majority of individuals

do not want editorials, or at least such opinionated ones, and on the other hand, quite a few said that's what they enjoy most about the newsletter. The majority said they have noticed and appreciated the improvement in layout and spelling. One individual did not like my old way of numbering pages, so as of March, that has already changed. I will cut down on editorials, but not eliminate them, ever.

Now, for some percentages. These figures reflect the status of those who responded, and all percentages reflect that number. 58% of my readers use an H8 and 42% an H89. This surprised me, as I thought more would be H89ers. Another figure which surprised me was the operating system chosen. I have been getting more and more mail asking for more CP/M material in this newsletter. 79% prefer HDOS, while 21% prefer CP/M. I like HDOS better myself, but am forced to go with CP/M more and more, as I know many of you are, to get the software I want. All my business software is running on CP/M, while my hobby and control end is under HDOS. I feel in many ways HDOS is better, but if you want to go business, or keep current, you'll eventually have to become involved with CP/M. That doesn't mean you have to know it, just how to load it and get your programs running. I'm taking this space here to tell you, instead of writing an editorial on it, since many have been asking my opinion. I don't want to start the HDOS vs CP/M war over again, so enough of that.

Of all the system users, only under 3% run nothing but the cassette system. Those using only the H9 terminal are about 1%.

Magazine and newsletter statistics. 94% of those responding subscribe to REMark, with it's average rating being 2.2, which is well above average. This is on a scale of 1 (excellent) to 5 (very bad). This is good news, since it's quite a bit higher than last years results. That means REMark is getting better and being more appreciated--growing with the times. Congratulations REMark staff. H-SCOOP and BUSS are running neck to neck just like last year. 86% of my readers receive BUSS. The average rating for this newsletter was 1.5, and BUSS was 1.7. Micro Media Magazine didn't have as close a following, with 20% subscribing to it. Micro Media is a disk based newsletter and more expensive because of this, perhaps that explains the low following. Their average rating was 3.4.

Now for the Non Heath Magazines. The following is in chart form for easier comparison.

BYTE	61%	2.6
MICROCOMPUTING	65%	2.0
CREATIVE COMP	30%	3.3
DR. DOBBS	20%	3.2
INTERFACE AGE	33%	2.8
PERSONAL COMP	38%	3.5
ON COMPUTING	23%	3.8

It's easy to see since Microcomputing (formerly KILOBAUD) is tops, and that's understandable since they have been carrying the most Heath equipment related articles. They therefore have the highest rating and the highest following. Hats off to you Microcomputing. I have long since discontinued Dr. Dobbs, and this seems to be the feeling of most. Interface Age was rated higher than I thought. That's it for the statistics, the more detailed items will follow in

future issues.

* I am receiving more and more requests for evaluation of business software. Since I do not have the time or money to evaluate everything out there, I'm asking for your help. If anyone is running specific business software and is either very pleased or displeased with it, please let me know. If you have the time to write a short report, your personal experiences with it, it would be greatly appreciated. I'd like to add a monthly REPORT column to this newsletter, but if the feedback is not there, this will not happen.

* Dean Gibson writes in to let us know that contrary to a statement made in last month's issue, 8080 source code is NOT compatible with 8086 and 8088 code. Dean does much assembly language programming on different CPU's, and has had experience on the 8086. Larry Reeve of Polybytes further comments that 8080 and 8088 are similar but not compatible.

* There has been lot of controversy on mounting a high density drive (like the Tandon TM-100-4) in the H89. It is said that the practice should not be followed because of interference problems with RFI and flyback voltages. The 80 track drives record tracks at a lower density than the 40 track, and are more sensitive to outside interference. I have had several users of the Tandons saying they have mounted them in the H89 with no problems. I even have one out in the field, that has thus far shown no problems. Al Rodemann sends in some information he found about the drives.

When he first mounted the drive in the H89 he noticed many seek errors and other problems and finally gave up. He noticed that the drive in the H89 was grounded to the power transformer by means of a metal bracket, which is actually used for mechanical support. Attached to this same location was the ground for the anode of the CRT. Al removed the metal bracket and substituted it for one of a non conducting material, placed the Tandon back in the H89 and grounded it to the CPU board. He reports no more problems or errors since implementing this procedure. He also says that the Teflon lubricant I mentioned a few issues back (TRI-FLON) does a great job in quieting noisy Tandons.

* Jim Texeria's UD.DVD all purpose driver is said to work OK for the H25 if CTS is set to YES. He has a new one for the H25, more next month.

* Update on the HUG CHICAGO CONVENTION from HUG headquarters. First let me correct a misprint in the last issue. Hotel accommodations are \$60 per night for a single, and \$70 for a double, not \$16 as reported. This was a misunderstanding on my part. Bob Ellerton gave me the following info.

Friday night and Saturday morning will be registration time. This will be a chance to walk around, see who's there, meet friends etc. Then the day will start with a talk from Heath president, Bill Johnson. This will be followed by a talk by Jerry Pearlman, financial VP of Zenith. Tom Dornback, VP software development for H/Z software and others will take the floor next, followed by Barry Watzman. Greg Chandler will present a talk on HDOS, and there may be a talk on Microsoft. All these will be followed by a question and answer period.

A general BS period will be held from 6-7, with dinner following, sponsored by Zenith Data Systems. ZDS will also be picking up the lunch tab which will be held from 12:30 to 2:00, with a keynote address by Don Moffett.

There will be door prize drawing for offers--big stuff!

Sunday till 1 PM will be free time, with a talk till 1:30 "Why HUG?". This will be followed by Doug Bonham, on

up and coming CAI, and there will be a surprise. 4PM will be closing time.

There is only space for 1000, and reservations are only made for that many. Any one interested, get your \$15 reservation fee per person (wife, family, friends all count extra) in to Jim Blake ASAP, once the 1000 reservations are sold, sorry, no more! They are budgeting about \$50 per person for this event. Your greatest expense will be lodging.

* You say you're looking for good business software? DBASE II is here. This is a full and powerful data base package, which makes your micro think it's a main frame, or at least a mini. Memory requirements for large data bases do not exist, because all file management is done on disk drive. This would make it extremely powerful for 400K Tandons, DD drives, and 8" systems.

What can you do with it? Define your records and fields any way you want, limit 65000 records. Index and search on ANY field. Easily set up (that's what I'm presently using it for) large mailing lists with capabilities such as enter names, delete, modify, sort, print all, some, print mailing labels, and much more. Check book programs are no problem. There are even examples in the comprehensive manual for a complete business system consisting of Payroll, Inventory, A/R, A/P, G/L and more.

Since it is very powerful and uses statements such as IF ELSE ENDIF, DO WHILE UNTIL, and handles mathematics, relational operators, logical operators and more, you can configure it and program it to do just about anything you want. The catch? There is none. You buy the package and receive 2 diskettes. One sealed and one not. You play around with the unsealed disk and see what it can do for you (limit 15 records). If you are not satisfied within 30 days, you return the package, with the sealed pack unopened, and receive a refund! What can be more fare? I feel this package, coupled with Spellbinder gives you just about all the power and capabilities for business use you will ever need. The price?

It lists for \$700. You can sometimes find it in the high \$500 range. But wait! This story is similar to that of Spellbinder. I will not get involved with dealership with any product if I do not personally know it and am pleased with it. Then, and only then, will I become a distributor so I can give YOU the best possible price and support. I was so impressed with DBASE II that I became a distributor and will offer this package for \$495! I know of no better deal, nor a better data base program. I was using Configurable Business System for awhile and was impressed with that, but swiftly outgrew it (after I entered many thousands of names only to have to re-enter them all in DBASEII) and then heard from subscribers of DBASEII. I purchased it and tried it, and dumped CBS. I'm not picking on CBS, but it's more expensive, and very limited, although it is easier to program. If this is what you were looking for, and want more info or want to order, call me.

* I received some feedback from Charles Bennett, perhaps the first Heath computer owner of an H89 to try the Percom DD disk controller board. First let me say although they advertise a board for the H89 and one for the H8, the H8 version doesn't exist. He said the SY device driver alone takes up 26 sectors of space. A Siemens type disk will give you 676 sectors free, while a Tandon DS DT will give you 2808 (which is about 718K). Charles says you can switch between using your drives as hard or soft sector (single or double density) with a RESET command from software, nothing to change in hardware as in others. It will support up to 4 drives in any combo of SD or DD. Charles said the DG disk management package works on this controller board.

Unfortunately, the parallel port uses 340, which is what the standard Heath serial board uses for LP, so if you use this built in parallel port on the disk controller board to drive an Epson or Centronics, you will lose the use of that serial port. If you want more info, call Percom toll free at 1-800-527-1222.

* Update from Bob Todd in reference to the **CP/M SIG/M** software available as mentioned on page 3 of last month's issue. Bob says SIG/M handles only 8" SS SD format. If you desire any other format, Bob is the guy to get in touch with. If you want to see what's available, send a SASE to Bob and request volume 051, which is the complete catalog for disks volume 1 through 51. In case you did not receive last month's issue, Bob's address is 1121 Briarwood/ Bensalem, PA 19020/ (215) 752-4604.

* **WILLIAM CAMPBELL, M.D./** 855 Smithbridge Road/ Glen Mills, PA 19342 writes to tell us that his "**Getting Started with HDOS and Assembly Language Programming**" course has had a demand greater than anticipated, so another printing was made. For those not familiar with this course, it is an aid to help the beginner to get started in Assembly language programming. The manual is good for beginners and will take you by the hand to help you get started. Doc's reputation speaks for itself. He has published many articles in REMARK. This primer is still available directly from William for \$15.

Doc also called to tell me that an **ASCII PS. PRINTWHEEL** is available for the **DIABLO** printer. It's a CUBIC 96, with a Diablo #311830-01. It is available from Guckes Brothers/ 7203 Browning Road/ Pennsauken, NJ 08109/ (609) 665-9044. I don't have an exact price, probably about \$65 since it's a 96 element wheel. I don't know if I care for the Cubic look, but it seems to be all that's available without special order of custom wheels, which would cost more.

* Anybody wishing the new update for **SPELLBINDER** should send the original distribution disk with \$50 to Lexisoft Inc/ Box 267/ Davis, CA 95616. They will send you the new version diskette with update sheets for your manual. There are also some additional special function keys used for this version. Those who use the special keytops will be able to obtain them in about a month. I'll let you know more when I get the information. I have a few sets on order already. If you are into assembly programming and miss the IOS to configure the system yourself, you noticed it is not on the diskette with the new release. It would no longer fit. If you want it, send me a blank disk and \$1 for postage and handling and I'll send you back a copy. Without a disk, \$4.

* I have become all too well aware that there is no decent **ADVERTISING** method available for vendors of Heath/ Zenith equipment to market their hardware and software. The newsletters are getting overburdened. The problem with newsletters is that they only reach their subscribers, and of that, most of it is buried in the newsletters. A typical 1/4 page add in BYTE costs about \$1500. Perhaps 5% of those readers may be Heath/ Zenith users, and of that maybe 3% will see and pay attention to the add, with as little 1% showing any interest. That's a lot of advertising dollars to pay for such a service. And what about the small guy who cannot even consider that kind of price?

I am going to offer a direct mail advertising service for all vendors for nominal costs. I propose a quarterly mail package going to many thousands of potential interests with blurps from all interested vendors. Since I can mail 2 oz mail bulk for about 12 cents each, and about 16 sheets can go into such a mailing, it can easily be seen how the costs will be divided into reasonable amounts. Although the small guy will not be left out, don't expect this to be free. Anyway you look at it, advertising is expensive, but this proposed method will probably net the most return

dollars per invested advertising dollar.

Any non H-SCOOP subscriber send in your name to be placed on the master mailing list at no cost to you. Vendors, if you are interested, let me know, I'm in the process of preparing vendor blurps to be shortly sent out.

* It appears that **TARCO** is headed for **very BAD GUYS** mention. Since there will be no such column this month, I will hold off till next month. Any feedback about them, send it in for evaluation. Any interested buyers of TARCO stuff, BEWARE.

NEW STUFF

Before I start I want to make a comment. In the future I will refuse to print any new stuff that does not include a phone number. If you are marketing something, you should be available by phone for product information and support. I am spending a good part of every day looking up vendor phone numbers and vendor information that I have in my files for individuals wanting support and product info. I feel like Grand Central Station information service! If you need info, call the vendor, not myself. He can answer any questions on his product far better than I can. If you have a problem, call the vendor, again, he can give you better info. If you want general information, or what to know who has what and what is what, then feel free to call me. Also, any vendors that 'do not supply satisfactory service' and support, please don't send me your **NEW STUFF**. If I receive enough feedback against a vendor, I will no longer accept his **STUFF** for printing.

Mean guy? Definitely not. I'm trying to keep this column reputable, and not print the junk and this is the start (and only the start!).

SOFTWARE

* Patrick McNally/ POB 578/ Haleiwa, HI 96712 has developed a **WORDSTAR** enhancement package for the H89, H8/H19 users. It upgrades the non-Heath 3.0 **WORDSTAR** by making use of all special function and keypad keys (AMEN to that, it's about time--that's what got me turned against WS and to SB!). I believe a future Heath release also supports special function keys. The original functions remain intact. Patches can be entered by merging a supplied .COM file under DDT or using the **INSTALL** utility supplied by WordStar.

It runs only on Heath CP/M 2.2.03, and is supplied on 5.25" hard sector SD diskette for \$19, which includes doc, or \$13 for doc and listing.

* Dave Brockman of **FBE RESEARCH CO/** Box 68234/ Seattle, WA 98168 sends in some updates (see Hardware). First off is the **SCREEN DUMP DRIVERS** for \$14.95. When the drivers are opened, they will dump to a printer anything on the H89 screen, including graphics, and reverse video for an Epson printer. Contact Dave of FBE for more info (phone # in hardware section).

* **XTRA SOFT/** 4910 Roman Dr/ Louisville, KY 40291/ (502) 499-1533 has available a **Small Business MAIL-BASE** software package which interfaces with their **POINT OF SALE/INVENTORY** business package. It is an advanced customer inquiry/ mailing-list system allowing the small businessman to maintain an easy to access customer data base. It is capable of keeping track of up to 32,000 records with name and address plus 15 flags, and room for comments. Customer records indexed by ZIP (I will do an article in the future on INDEXES in DATABASES--very important, if feel, and much neglected). Selected records can be printed on any form of mailing labels, index and Rolodex cards, user defined forms, etc., or merged into form letters for mass

mailings.

* Update and new software from Rick Kerble of **SUNFLOWER SOFTWARE/** 13915 Midland Ave/ Shawnee, KS 66216/ (913) 631-1333. Please note that this is their **NEW ADDRESS AND PHONE NUMBER**. They were previously 6304 W 76th St/ Prairie Village, KS 66204. They have a new product called **S-BASIC**, which is a structured BASIC language and translator by Wm Clarkson for the H/Z computers with HDOS. S-BASIC retains the merits of BASIC while replacing its outdated structure with a structured programming technique. It can translate the finished product in either Extended BH BASIC or MBASIC. Price is \$49.95 which includes both translator versions. These and many other fine programs from Sunflower (such as WORDPRO and HDOS professional word processor, SPELGUD to proofread words against its dictionary, DATA PROCESSING files, APPLICATION software, EDUCATIONAL, UTILITIES and more) are available from **SUNFLOWER** or most Heathkit centers. Write for a full catalog.

* A place I haven't heard about before, but sounds good, is **HUSKER SYSTEMS/** 4517 N. 61st St/ Omaha, NE 68104. They are a recently formed software company specializing in Heath/Zenith Systems, formed of data processing professionals. For a complete catalog, write them, but here are the highlights. (I don't know, the doc never said, but I assume this is all for HDOS)

COMPUTEXT, for \$24.95 is a Word processing system featuring right hand justification, wrap-around entry and more. **WORD WIZARD** for \$44.95 is an expanded version of **COMPUTEXT** which includes graphics capabilities when used with the MX-80, and includes additional capabilities. (Owners of Computext can upgrade for \$20). **ASSEMBLER TOOL KIT** is a collection of 20 ACM assembly sub-routines which can be X-TEXT'd into an assembly program. Price is \$19.95. **DISK*HELPERS** for \$24.95 is a disk of 3 programs which include **DISKDUMP** to enable the user to view and alter disk sectors, **LIBRARY** which reads a collection of disks directories and prints a library listing, and **DISKSTAT** which replaces the lost HDOS 2.0 disk stat commands and allow you to write them to a printer. **SYS*HELPERS** priced at \$39.95, contains all of **DISK*HELPERS** programs and more; including a comprehensive memory test, and all ACM files found in **ASSEMBLER TOOL KIT**, plus a special device driver.

Also is **NAVY MARS SYSTEM** which is designed to operate with the DFD Systems RTTY package, or stand alone mode. Provides full function of a Mars Station. Priced at \$22.95. **GAMETIME '82** for \$22.95 is a collection of 20 popular games with high usage of Graphics, all linked together by a menu. **GAMETIME '83** for \$34.95 contains 30 games including MBASIC and ABS types. **AMATEUR RADIO STATION** at \$29.95 automates any HAM station. It includes Station Logging, Log search/ inquiry, references for "Q" signals, Antenna measurements and more.

* George Schreyer/ 412 N Maria Av/ Redondo Bch, CA 90277 announces that he is distributing the **USUS** (UCSD System Users Society) Software for the H89. If you are a USUS member, you can order USUS library software for \$17 per volume (3 disks per volume) in Heath Hard sector format. There are presently 8 volumes and more on the way. If you do not belong to USUS, you can for \$20 annually, which includes a quarterly newsletter. This is available from USUS, Chip Chapin Secretary/ POB 1148/ La Jolla, CA 92038.

* From **EVERYWARE/** POB 60802/ Sunnyvale, CA 94088 (No phone #) comes some new software, both interactive fiction story telling using your computer. **SIX MICRO STORIES** is a program where you take part in six short stories, each with many outcomes. What you do will

determine the outcome of the story. Price is \$17.50. **DRAGONS OF HONG KONG** begins as you walk into Big Al's bar to meet professor Goodman who has just discovered a terrible secret. The secret takes you to the Far East where you have a chance to free the world of an age old blight, clear your name of a despicable crime, and finish the story in the arms of the woman of your dreams. \$21.50

* New from **ULTIMETH CORP/** 24025 Fernlake Dr/ Harbor City, CA 90710/ (213) 539-4276-9AM to noon (the 400K software SY people). Both of these programs are for the Magnolia Microsystems extended memory (128K-see last month's issue) board for the H89, under HDOS 2.0. **AX.DVD** is a device driver allowing user to treat the extended memory board as a single disk drive holding 111K bytes. **CACHE.ABS** is a program supporting auto "cache memory" capability. When the program is run, it installs a facility which monitors and references to the system disk device driver (SY.DVD) and maintains a duplicate copy in "cache memory" of the most frequent disk sectors referenced. A disk read is satisfied from "cache memory" whenever possible; a disk write is duplicated to "cache memory". The price of this package is \$40 from UltiMeth Corp. This includes doc and shipping to US addresses. Supplied on HDOS 5" SS 40TK hard sector disk.

* Burt Hulland of **HULLAND ENGINEERING/** Room 217/ 555 Broadhollow Rd/ Melville, NY 11747/ (516) 420-0970 says a new version of his HDOS is in the makes and should be released by the time you read this. This version **HDOS A.0**, includes a number of improvements suggested by users of the 0.1 version, as well as his own ideas. Some major improvements include the supporting any disk drive that HDOS 2.00 can, but a hierarchy of directories on one disk. This system will be larger than the 0.1 version, but still much smaller than HDOS 2.0. For more info, refer to back issues of this newsletter, or contact Burt.

* More software releases from **HOYLE and HOYLE SOFTWARE/** 716 S. Elam Av/ Greensboro, NC 27403/ (919) 378-1050. **BRICK BREAK** for \$16.95 gives you a wall of color, five balls, one paddle, and three choices of speeds. You try to knock out all the bricks in the first wall and you get another free. Super features. Includes source code and all routines which will run on the new H89 color graphics board (stay tuned to H-COOP for more). Uses keyboard. Runs on HDOS with 32K using the H8 and color graphics board.

RAPID RECALL for \$16.95 is a visual musical version of Simon Says. You attempt to repeat all 40 notes the computer randomly selects (it also includes color, sound, number clues). Of course they still have their popular **Galactic Experience** for \$18.95 and their newest hit **A Remarkable Experience** for \$18.95. These are available from most Heath stores, or through Hoyle. They accept Bank cards, COD's, or checks, and do take phone orders.

* **THE SOFTWARE TOOLWORKS/** 14478 Glorietta Dr/ Sherman Oaks, CA 91423/ (213) 986-4885 has a new **C COMPILER** which boasts improved performance and features. It supports full C data structures, pointers, multi dimensioned arrays and more. These features make it highly competitive with compilers selling at many times its price. Price is \$49.95 plus \$3 shipping (\$2 for 5" disks), available in 5" Heath/Zenith (CP/M or HDOS) or 8" CP/M.

* More from Dave Powers of **GENERIC SOFTWARE/** POB 1154/ Troy, MI 48099/ (313) 879-6903 for users of H8-H19 or H89 operating with HDOS or CP/M and a single disk drive. **ENERGY MISER** is a Visi-Calc type program for maintaining and displaying energy usage and cost data. A useful analysis tool when dealing with volatile nature of energy costs. Provides for hard copy output of summary reports, bar plots of energy costs, and

OFFERS

Suppliment to April 1982 H-SCOOP

The re-organization of the H-SCOOP disk offers is now complete. You asked for a complete descriptive listing, you got it! These offers replace all past offers, which no longer exist. All monthly offers (HS-XXX code) are made available as-is. Every effort was made to ensure compatibility with the present HDOS operating system, and all known bugs have been removed. The disks are for the most part quite filled up. I know of no other software offers where you can get so much for so little! Non subscribers of H-SCOOP add 50% per disk. Please order all monthly offers by code number. All prices include shipping, handling, and 5" 10 hole hard sector disk. NOTE: The first disks shipped will be reclaimed from my large stock of offers left over before my re-organization effort. They have only been copied to once, and are warrantied to be as good as new. These must be used first before new disks will be used. Remove this from your newsletter and keep it as a handy reference to what is presently available. More will be added and announced in H-SCOOP as they become available.

HS-001 GENERAL UTILITIES & TOYS \$10

This disk contains the best of some of the utilities from previous disks such as; BAUD (ABS,ASM,DOC) to set the terminal/computer baud rate from the HDOS prompt, CHGLABEL and CHGNUMB to change the label and number of a disk without re-initing, COREDUMP to dump contents of memory, DEV to display current devices in use, DISKERR.BAS to give you disk error statistics from a BASIC program, INFO to show how your console is set up, LISTFILE.MB, MAP which allows you to get a cross reference table of variables for your BH BASIC programs, MURPHY'S LAW documentation, R.ABS for SY1: RESET/DISMOUNT/MOUNT one key command, SCREEN.BAS to make screen formatting easier and make use of special function keys, VOXDUMP for the H8 to make it talk, and some MBASIC games--Blackjack and Crazy 8. H19DEM.MB an excellent terminal demo program with an excellent ending (a talking face appears). 52 sectors free

HS-002 WORD PROCESSORS & EDITORS \$10

This disk contains two word processing systems which operate under MBASIC; EDITOR and DOCUFORM, complete with documentation. Good value for the price, supports features such as right justification, formatting, etc. 112 Sectors free.

HS-003 H19 GRAPHICS & PRINTER PICTURES I \$10

Contains many Stupid Fascination Programs. These run under MB and do nothing but print vivid non ending graphics to the H19 screen, some in kaleidoscope fashion. Good way to learn graphics programming. Includes BOUNCE, FIREFLY, SCOPE1, SCOPE2, SCOPE3, SNOW, and SPIRAL.SFP's. Contains other graphics displays such as WORM.MB, KALEIDO.MB, MAZE.MB, MAZE1.MB, and graphics games such as PINBALL.ABS, DOGRACE.BAS, TTTOE.BAS (Tic Tac Toe). H19GRAPH.ABS is a screen ABS graphics program for the H19, and SCREEN.ABS is another H19 screen graphic program, both to check out the terminal and screen. Pictures to output directly to a line printer include FROSTY, JFK, MERRY, PUNT, and SANTA. Also contains a maze generator for hardcopy use. A Treasure of pictures and graphics. 18 Sectors free.

HS-004 H19 GRAPHICS & PRINTER PICTURES II \$10

Another treasure of screen graphics and pictures is found in this volume. Graphics games include BATL.MB (Battleship), JUMP.MB, SHOOT.MB, and OTHELLO.MB. BIGTYPE.BAS allows you to draw large characters or messages on a printer. DAISEY.MB is a graphic program you wouldn't want to miss, great motion graphics. Of course, more Stupid Fascination Programs, including CHAIN, HEX, and WORM. Another excellent graphic program is TOWER.MB with great moving graphics for you to play against the computer. DRAW.MB is a program allowing you to draw things on your screen. GRAPH.DOC is for hardcopy listing which shows the graphics characters of the H19 in the sequence most likely to be used. More printer picture, SNOOPY.PIC and TWEETY for hardcopy print. 18 Sectors free.

HS-005 MISC BASIC: LABELS, RANDOM FILES, PLANT DATA BASE \$10

This is a collection of odds and ends BASIC and MBASIC programs including a mailing labels set of programs in BASIC, a BIORYTHM.MB program for charting your (or someone else's--good money maker) biorythm, a RANDOM ACCESS data base set of programs which is said to be worth more than the disk price alone--This allows you to create random files to your specs without knowing anything about random file I/O (under MBASIC). You can add, change, delete, print, and list anything in the file; a PLANT DATABASE program which contains information and answers about many types of plants, such as sunlight and water requirements (97 sector data base), a MBASIC QUIZ program and a Vici-Calc type TRIANGLE.BAS program to calculate unknown values of a right angle triangle. 6 sectors free.

HS-006 CHALLENGER \$10

This disk contains a collection of 2 programs. The famous ATOM20 variety adventure type program is found here (in BASIC) where you must survive after a nuclear holocaust. OREGON TRAIL (BASIC) is a travel adventure game going down the Oregon Trail. 46 sectors free.

HS-007 HDOS 2.0 PATCHES \$10

This is the basically the same disk as the May '81 offer only with some corrections, improvements and additions. Contains over 100 sectors of documentation on applying different patches to HDOS 2.0 and even some for use with SYSMOD. These include patches to patch PATCH, Unlock FLAGS patch, Eliminate irritating FLAGS message, Reduce motor spin time mod, Fast boot mod to bypass ACTION BOOT message, CLIST and SUBMIT mods to allow them to work under 2.0, DATE with no CR mod, NO DATE mod, 4 sector directory mod, speed up DATE command, REMOVE STATUS LF so all fits on screen, DEBugged DBUG mod to allow it to work properly, FIX STATUS report so it prints better info, 0-org mod for 1.6, fix to REMARK issue #12 MOUNTALL, more space in ERRORMSG.SYS file, mods to ASSEMBLER and XREF for HEX output, several mods for SYSMOD and much more. Actual patched files include a modified HEX ASM.ABS, modified ERRORMSG.SYS to take only 7 sectors, modified PATCH program to allow you to patch files without knowing ID number, FLAGS which allows you to UNLOCK flags, SUBMIT to work under 2.0, many ACM files, disk statistics (modified STAT program) for HDOS 2.0 and more. 22 sectors free.

HS-008 QUIKDATA SELECTRIC AND MISC DRIVERS \$10

This disk contains the original selectric driver that goes along with my hardware article on how to drive a selectric with a parallel port. Includes source code. Also contains a version for the serial card with SET options. Diablo 630 driver to allow bi-directional printing from HDOS, a driver for Anderson Jacobson terminals, a H25 device driver, and associated ACM's. 198 sectors free.

HS-009 MISCELLANEOUS BASIC PROGRAMS I \$10

Didn't know where to stick these, and they were much too good to toss out, so I stuck them on one disk. A collection of BASIC and MBASIC programs. The BASIC one includes the very long game (?) of SWARMS, where killer bees are trying to attack and you must try to stop them. A game could take more than 15 hours, a long playing and exciting game. The rest are MBASIC, including; DATE-you enter a date and it tells what day it fell on, HANG (a graphics HANGMAN game), MATHSQ -a magic square generator, OPAMP1, OPAMP2 and OPAMP3 for designing of OP AMP circuits (uses pseudo graphics for drawing a basic circuit schematic of each circuit type. You type in a few parameters and it generates all components required), which include low and high pass filter, bandpass and notch circuit, and various amplifiers including inverting and non inverting, SINK

(another graphics one) where you must sink a ship, STACAP-an improved state capitol program which gives you hints. As a bonus, DIET.MB is included. It is a health related program which calculates your daily calorie output depending on your daily activities, and calculates how many calories per day you must loose over any specified period of time depending on weight, sex, height. 14 sectors free

HS-010 MULTI KEY SORT ROUTINES \$6

A very popular sorting routine written in "C" with source code included to allow you to sort any fixed field file of records on any field. For this price it is supplied on once used but guaranteed disk. A very fast and versatile sort program which equals or betters many on the market costing lots more buck\$. Can sort a file in memory on ANY key. Executable fast ABS code which runs on a 1 drive system and takes up only 16K with HDOS. Sort up to 25 fields, any length, ascending or descending order. Full documentation on the disk. 118 sectors free.

HS-011 FROM THE UNIVERSITY OF HOUSTON VICTORIA CAMPUS \$8

This was sent in from students at the University of Houston and contains many good programs, my favorite is BIGFONT.BAS. You can have large letters or messages printed on the screen of your H19 in the same fashion as if the human hand were actually writing the letters! Very unique and exciting. Well worth more than the price of the entire disk! Other files included several MUNCHKIN DATA files to be used with the MUNCHKIN program from Software Toolworks, CAKE and CHAIR.ASC, along with full documentation, which are MBASIC programs of simple Computer Fantasy Simulation for kids, and a multi player, non graphic NIM game FIBNIM. 16 sectors free.

HS-013 MULTIPURPOSE DUP & DUMP, AND H8 MISC ROUTINES \$10

This was taken, modified and enhanced from my 10/81 offer, MIXED BAG, which was then the 2/82 offer. This is simply a renaming of the 2/82 offer, so if you have that, you won't need this. This disk contains DUMP and DUP programs that can be used on any 5" disk from 100K to 400K and will make all other DUMPS and DUPS no longer needed. These will handle it all and then some. Very versatile, and contains some outstanding features. DUP two 100K or 400K disks from any drive # to any drive # (must be the same types drives, of course). Can even DUP CP/M disks under HDOS. The DUMP also works on standard 100K or 400K disks. Want to get the most of your H8? Various routines are included to really make your H8 front panel display come to life. Complete doc included, 0 sectors free.

OTHER THINGS FROM QUIDATA INC.

Besides these monthly software offers, QUIKDATA INC, offers various other products, hardware and software. This is the more expensive software and hardware, but you won't beat it for these prices and still get full support. I never sell anything I do not fully believe in myself, and use myself, thus am able to make honest recommendations and offer full support.

SPELLBINDER WORD PROCESSOR AND OFFICE MANAGEMENT PACKAGE \$295.

Compare this price at the list of \$499 and then compare to any other word processing package out there and this is the way you'll go. Makes use of the special function keys of the H89 (there's even color coded keys available to replace the H89 function keys which are labeled just for SPELLBINDER). Supports full proportional spacing features of a Diablo type printer for true printshop typeset quality (what this and H-SCOOP is prepared on). Offers all the full features of a good word processing system and many more. Can do two column print, mail merge, Zip sort, Sort on any field, multiple un-attended copies. . . Interfaces with THE WORD (\$75) for complete spell check capabilities. And it's easy to learn and easy to use. You don't have 1001 control codes to memorize and figure out. Comes on H/Z 5" CP/M SS SD disk.

DBASE II \$495

Compare with the list at \$700 and compare the features with any other micro data base program on the market. Fast, handles large files of up to 65000 records (limited by disk not memory), sort and index easily and fast on ANY FIELD, expand record length at any time, full report handling, full programability (IF THEN ELSE, DO WHILE etc). A cinch to make a great big master mailing list data base, inventory, student records, payroll, A/R, A/P, G/L. This one can do it all! Perhaps this program along with Spellbinder is all the business office will ever need!. Supplied on Zenith 5" CP/M or CP/M 8" disk.

TANDON 5" DRIVES

TM-100-4 DS DT DD 3 ms step rate \$395
TM-100-2 DS ST DD 5 ms step rate \$300
TM-100-1 SS ST SD (Replaces Siemens types) 5 ms step rate \$225

MAGNOLIA MICROSYSTEMS H89 DD SETUP \$495

Compare at list of \$595, includes H89 double density board which will handle up to 4-5" drives, AND 4-8" drives plus your standard controller for 3 more 5" drives. Comes complete with cables for 5" and 8" and CP/M operating system and all else needed. HDOS driver also available with this for H37 and H47 compatibility. A much more versatile and flexible way to go than the H37 and H47 and a better price.

. . . AND DRIVERS GALORE!

DKH17V2.DVD HDOS 400K driver for use with single density H17 type controller card to give the Tandon TM-100-4 400K storage under HDOS 2.0 with no modifications needed to the system. \$40.

DKDDFLP.DVD HDOS driver for Magnolia Double Density card \$40.

BIOS80 for CP/M 2.2.02--allows the Tandon TM-100-4 to be used with the Heath 2.2.02 CP/M for 392K storage per drive. \$30

BIOS80 for CP/M 2.2.03--allows the Tandon TM-100-4 to be used with the Heath 2.2.03 CP/M for 392K storage per drive \$35.

MORE COMING ALL THE TIME. For more info on any of these write or call for seperate info. Stay tuned to H-SCOOP, your contact with the future, for more info and items.

inflation rate calculations. Supplied as a binary program for fast execution. Requires 40K memory, price is \$24.95.

WORLD MAP for \$19.95 requires 48K HDOS or 56K CP/M and requires MBASIC interpreter. It's an introductory course in world geography. The program explains the basic entities of our world. A presentation of the major continents and oceans is enhanced by graphics. After the names and locations of the continents are presented, a review of the subject material is given. **WORLD MAP** has an appeal for the entire family. Both products shipped on 5" hard sector disks. Add \$2/order for shipping and handling, and specify operating system.

* Four new UCSD PASCAL programs are available from Christopher Simmons of **COMPUTER SYSTEMS SOFTWARE/** 2802 N Greenwood St/ La Grande, OR 97850. These are all supplied on 5" hard sector disk. **XCATALOG** for \$20 is a directory catalog of programs which reads Pascal disks and prints a listing in alphabetical order the file name along with the type of file and the disk that it is located on. **XREF** for \$20 is a cross reference program that reads a Pascal textfile or program text and lists the terms and line numbers where the term is located. **LIST** for \$10 is a program that reads a Pascal program text file and formats the output to the printer using title, subtitle, and utilizes pagination. **PHONEME** for \$5 is a small phoneme dictionary that can be built by adding a new word and phoneme equivalent, change a phoneme spelling, find the phoneme for a particular word, delete any word, and phoneme or list the entire dictionary to the printer. Uses a disk for data files allowing a max of 75 words per letter of the alphabet. All supplied for H19, but if sent **CLEAR SCREEN** and **ERASE END OF LINE** they can be configured for any terminal.

* **GREGORY MARSH/** 5279 Miles Ct/ Woodbridge, VA 22193/ (703) 590-3360 announces the newest release of **NEWSTAT**. **NEWSTAT** is a utility for HDOS that combines the **STAT** functions for version 1.6 and 2.0 of HDOS. This version supports features of the **UltiMeth** (400K) driver. The step times for each drive (up to 6) are displayed along with the last volume mounted and track accessed. For a hardcopy output of what you get, plus a complete catalog of other products, send a **SASE** to Gregory. This package is priced at \$15.

* New from Tom Jorgenson of **SOFTWARE WIZARDRY, INC/** 122 Yankee Dr/ St. Charles, MO 63301/ (314) 946-1968 is a very different **MODEM** utility. It is a highly intelligent **MODEM** package for the H8 and H89 HDOS system called **ZLYNK**. It is the first HDOS modem utility to support features such as: Screen-oriented and user-friendly, self configurable, Handles **XON/XOFF** protocol (same as **CTRL S/Q**), Supports the Christiansen **XMODEM** protocol as used on CP/M dial-ups, Supports dumb terminal protocol, Transfers between users or a user and a dial-up system are handled via any of 3 protocols, **RESET**, **DISMOUNT**, **MOUNT**, **CAT**, **DELETIONS**, and more supported without ever leaving **ZLYNK**, Contains a Real Time Clock and connect time clock, Maintains multiple connect time logs, Allows auto login and dialing, Supports screen dump to disk or printer with built-in edit function, Handles baud from 110 to 4800, Built in Help, and much much more. Surprisingly the price is only \$24.95. For more info, or to order, contact Software Wizardry. This looks like a top notch product, the **Modem** program to end all **Modem** programs!

* **PEARL III** is available for about \$650 from ZDS distributors, for the Z89 and Z90 type computers. It is a tool for system integrators and experienced programmers, permitting them to develop software for custom applications. The idea behind this is to allow software designers to spend their time doing analysis of customer needs rather than spend their time programming. **PEARL III** operates under CP/M and requires 56K, and **CBASIC**, and is supplied on 5" or 8"

disks.

* Those of you following my Pascal corner in **REMark** know by now that the **Lucidata Pascal** is the one I prefer, and is the best value for the price. Well, it just got better! **Larry Reeve** of **POLYBYTES/** 325 19th St. S.E./ Cedar Rapids, IA 52403 has just released the latest 3.8 version of **LUCIDATA PASCAL**. It is available either under HDOS and CP/M. Refer to **REMark** or past issues of this newsletter to find out why this one is better for HDOS than any other.

Hardware requirements are Two SD 5" drives, 32K memory with 56K recommended, HDOS 1.5 or higher, or CP/M 2.0 or higher. Many new features have been incorporated into this release, along with those present in the past release we have grown to respect. Updates for 2.8 version owners are \$60, no disk returns required. Price for new buyers is \$145. Write or call **Larry** for group rates and discount prices. Phone (319) 366-3077 6-9 PM C.S.T. Disk media are 5" hard sector H17 format, 5" DD H37 format for HDOS and CP/M and 8" for CP/M only.

* The Original Adventure Game is now available from **Walt Bilofsky** of **THE SOFTWARE TOOLWORKS** (Address in this section). Do you dare enter the dangers of **Colossal Cave**? You are the adventurer seeking fabulous treasures, mythical adversaries to overcome, and initiation into the secret spells which operate within the cave. If you can solve all the puzzles of the cave, you are entitled to a free serial numbered **CERTIFICATE OF WIZARDNESS**, suitable for framing (Looks real good, with Gold seal and all!--ed).

The original Adventure program has inspired hundreds of computer game creations. Now you can experience for yourself all the wonders of the original. Requires 48K, 5" HDOS or CP/M hard or soft sector, or 8" CP/M. Order #222, price is \$19.95, add \$2 per order shipping and handling (\$3 for 8"). Call or write for their complete catalog of over 20 time proven programs.

* **MICRO MEDIA MAGAZINE/** POB 402286/ Garland, TX 75040 is making many software products available, which are of exceptional quality. They have a **PERSONAL ACCOUNTS PAYABLE SYSTEM** for \$24.95 designed to maintain information on your household bills. Many options such as **LIST**, **ADD**, **DELETE**, **UPDATE** accounts, **ENTER** bill, **PAY** bill and more are available. Maintains year-to-date totals.

STARTREK for \$19.95 comes on a dual format CP/M HDOS disk and requires 56K and **MBASIC** interpreter. It is an interstellar combat game with complete vivid graphics from **CyberSoft**. They are also handling **CAI** (Computer Aided Instruction) from **CyberSoft** for \$25, which includes a variety of math and reading and spelling drill programs. A program is also included to let you write your own **CAI** courseware.

In addition to these, they handle some of the programs from **EVERYWARE**, such as **GALACTIC WARRIOR**, some of the programs from **THE SOFTWARE TOOLWORKS** such as **INTRODUCTION TO BASIC PROGRAMMING**, **SPELL**, **ELIZA**, **ED-A-SKETCH**, a well known program **SYSMOD2**, and more. Micro Media Magazine, is the disk orientated Heath/ Zenith newsletter available for \$12.95 per single issue, or \$60/year. Their issue 5, Volume II includes games: **TEEN** which is a 4X4 screen grid layout where you have to move numbers to get them in the proper order, **CONCENTRATION**, **IMPOSSIBLE**, a checkers type of game, and more. For info on any of this, contact MMM.

* Rick Lutowski of **MICROTRAN Co/ 76 Flintwell Way/ San Jose, CA 95138/ (408) 226-4122** is a company specializing in **FORTRAN SOFTWARE** for the Heath/ Zenith computers. They feel that by concentrating on a specific language for a specific system they can better provide reliable software, compatible products, and best customer support. They have software out in three categories; **FORTRAN** callable subroutine packages, Stand-alone utilities, and Application software.

* Their **FORTRAN** callable Overlay package for \$22 consists of **FORTRAN** callable subs, **LINK** and a stand alone utility, **DC80**. Together **LINK** and **DC80** implement a simple, highly effective overlay capability of the 'chaining' variety. It contains many other features as well. The **FORTRAN** callable **AM9511 Arithmetic Library** for \$26 is another one of their programs. **MICRO-CORE GRAPHICS PACKAGE** for \$49 and **MICRO-CORE Device driver** for Percom Electric Crayon for \$23 are their **FORTRAN** graphics package disks. For complete information on these and more of their **FORTRAN** programs, contact **MICROTRAN**. All orders add 50 cents shipping per disk.

HARDWARE

* Dave Brockman of **FBE RESEARCH CO** (See Software for more info, and address), sells parallel interfaces for the H89 which are good for Centronics type printers (including Epson). They also have **MX-80 BIT IMAGE GRAPHICS** which is the Grafrax 80 option for doing graphics with the MX-80 (Dave sent me a sample graph done with this--very impressive!) for \$75. Dave says because of the bit graphics and the speed, it will not work with an 8141 type serial interface because it is too slow. It will work with FBE's parallel port though. Something FBE was doing awhile back and dropped has now emerged again. It is a **16K Expanded memory** for the H89. Called the **H89RAM** it sells for \$40 without memory chips, and \$55 with, assembled and tested. Dave says it's not very pretty, but it works.

The big news at FBE is the **H88UTI Utility Card**. It replaces the Serial I/O card and is compatible with Magnolia Microsystems products. It provides a calendar clock with battery backup, a bi-directional parallel port (with MX-80 printer cable and driver available), an optional math processor chip (**AM9511A**), and two standard serial ports (330, 340Q). Price is \$150. For that you get the clock and parallel interface all installed, serial port chips (you can steal that from your Heath board) are extra if needed. Phone number, (206) 655-3711 days, or 246-9815 nights.

* **RAY ALBREKTSON/ 619 W Dover Dr/ San Bernardino, CA 92407** is offering the last of his H89 **AUTO BOOTER** kits complete for \$11 postpaid. These are ideal for owners of older H89's who desire auto-booting on power up. Auto Boot is ideal for security or control applications. If you have a power outage and your system dies, you'll want it to come up again ASAP, without user intervention!

* Available at the end of April is the **H89 BUSS I/O**

EXPANSION BOX for the H/Z89 computers. Called the **M89**, the box has 9 slots which are compatible to those on the right hand side of the H89, and has many desirable features. It has a heavy duty power supply, cooling fan, AC line filter, and replaceable front panel, to name a few. 20 RS232 mounting positions and 8 larger cuts for 50 conductor ribbon cable are available on the side panel, along with 2 AC remote outlets plus 2 Molex type 12 pin connectors on the back panel. Microflash is presently working on a slave controller board as well as X-Y plotter, Voice synthesis, etc. Don't have enough room anymore in the H89? This sounds like the answer. All orders received before March 31, will qualify for a 10% discount. Allow 4 weeks delivery.

The M89 is available from **MICROFLASH CO/ 4916 B Carol/ Skokie, IL 60077/ (312) 677-4928**, in many different configurations. The entire box assembled and tested is \$395, the kit for \$315 (add \$10 shipping & Handling). They have bare board sets, cabinets only, parts sets, manuals only, relay sets, decoders, and more. Write or call to order or for more info.

MISCELLANEOUS

Last month I mentioned the Smith Corona low speed daisy printer for about \$850. Now I have one better. It's the Adler 1010 **ADVANCED ELECTRONIC TYPEWRITER** capable of computer interface for print. This is an electronic daisy wheel typewriter with many features, which lists for about \$1300. As a typewriter it is capable of 10 pitch, 12, 15, and Proportional Space, with various line feed sizes. It has a self contained 132 character two line correcting memory (with correcting tape). Featuring microprocessor technology, it's a very quite machine, with a print rate of about 17 cps. Uses plastic daisy wheels and ribbon cartridges. If you need a typewriter, this may be the way to go. If you need a typewriter AND a letter quality computer printer, and for a low cost, this probably is the way to go.

There is a serial printer RS-232 interface available for about \$350. Now for the rest of the story. If you are interested, write me. I'll saw this machine and was fairly impressed with the quality/ price ratio and will be handling this unit if the response is good enough. I'll be able to sell the typewriter for around \$1000 (which of course includes shipping and handling charges) and make the serial interface available for about \$300 which includes software driver. So it looks like about \$1300 for a daisy wheel printer and typewriter. If you are genuinely interested, contact me.

REQUESTS

* Richard Melvin/ 2103 First St/ La Grande, OR 97850 is looking for any information you may have available on converting the serial output of the H8-4 board to a parallel output. He is presently using a 74LS90, 74LS164, and 74LS374 but is unable to get the strobe needed from the H8-4 card. He is also interested in any information on going parallel to serial.

* Ted Morton of **Agricultural-Specialties Co/ 4867 Chambersburg Rd/ Dayton, OH 45424/ (513) 236-3583** is looking for hardware and software that will allow the use of an H89 as a TWX or TELEX terminal to communicate with other TWX or TELEX customers.

* Tor Tverre/ Monradsgate 3 A/ Oslo 5/ Norway owns an IDS 560 printer which is capable of graphics printing. He wants to know if anyone has a driver, or any other means to allow this printer to print the H19 graphics set. He is also wondering if anyone has worked on interfacing this printer to WordStar or SpellBinder to utilize the printers proportional spacing option.

* Jan Johnsen/ POB 993/ Bellevue, WA 98009 would like to know if anybody has ever got AUTOSCRIBE to work with HDOS 2.0.

WHO'S WHO

* Fred Freland, MNET 70705,121 has a module of code which will allow the selectric using my hardware circuit plans, to operate with CP/M 2.2.02. It goes where the old TTY physical routine was in the BIOS. This will allow an H8-4 to drive the selectric under CP/M. That's all I have. Anybody knowing more info on Fred's address or phone number, please contact me. Meanwhile if anyone wants the mod, and gets on MNET, contact Fred there. This was yet another misuse of micronet! Had I not happened to be on that night to glean information, this would have been gone to the bit bucket forever! Anything useful like this, please send it in for posterity, if for no other reason.

* Richard Melvin points out that the long motor spin time on the disk drives when using the Heath UCSD Pascal can be cut down the same way as was done for HDOS from the H8 front panel; change data at memory location from 040.110 from 170 to the desired spin time (e.g. 10 for a motor time of about 5 seconds). If anyone knows where this is on the UCSD system disk for patching, he'd like that information.

* Pacific Northwest HUG (PNHUG) has designated Nathan Hall/ POB 993/ Bellevue, WA 98009/ (206) 363-3927 as their contact person. Please contact Nathan for matters relating to the group.

TECHNICAL FORUM

SYSTEM CRASHES--NOW WHAT?

There have been many inquiries about system crashes and what to do about them. I will outline a few items to check which may be so simple they are often overlooked. System crashes and problems can be hardware or software related. I will deal with some points in this article about hardware related.

If your system goes down and dies for good, you most likely have a component failure. It is beyond the scope of this article to cover trouble-shooting procedures related to these cases. More often, as I have found, there may be intermittent problems. Symptoms may be occasional crashes, system will boot one time and not another, video problems that seem to come and go, etc.

The first thing I suggest, which I have written up previously, is to install a line filter on your equipment. No computer equipment should be operated without a line filter. Power line glitches and spikes can enter the equipment and actually write false data into memory. If it happens in the "right" spot, the system may die. There are many types of line filters available. I have a Radio Shack (yes, they do sell some good things) line filter multi-outlet box on one of my systems. On another I have a GESP-752 voltage spike protector which is a small device you plug into an outlet, and the computer equipment plugs into this. The latest Radio Shack catalog had this device listed under their own name. Another I am using is the KALGLO line filter. Keyboard Studio handles this one. Also note that you should never operate your equipment without a grounded receptacle (3 prong). This will help suppress RFI and properly protect your equipment. And while on the subject of spikes, always unplug your equipment during a lightning storm. I know of an individual who blew out an H8 which was plugged in, but not even turned on, during a lightning storm. Those kinds of voltage spikes will jump the switch contacts without any problems.

Along this same line of thought is static electricity. This can present major problems for computers. A static buildup or discharge can not only wipe out memory cells, but permanently damage some chips. Avoid static at all costs. The best procedure is to have proper humidity present to eliminate static. If this is not possible, get an anti static mat installed by your equipment. The humidity approach is best since constant humidity and temperatures is best for all equipment, especially diskettes, which are very vulnerable to temperature and humidity changes.

Another thing I have found is socketed IC's may become loose or build enough resistance in the IC pin-to-socket contact to cause problems. This seems to be worse depending on your environment, i.e., humid salty conditions being much worse than a dry non salty atmosphere. What to do in an extreme case is to remove the IC's, spray the socket with contact cleaner, and replace the IC, taking care to insert it and push it all the way down in the socket. No computerist should be without contact cleaner. For non extreme cases, I find it is usually satisfactory to simply place the PC boards on a flat surface (I use a carpeted work bench) and firmly press down the IC's in the sockets. The ever so slight movement will usually cause them to seat better.

Along similar lines, all interconnect cables should be suspect. I recently had a weird problem with my H89 (from H19 conversion kit). Suddenly there would be a click sound (like a circuit breaker) and the screen would flash, a bright vertical line would appear down the center of the screen, and nothing. I'd turn the thing off, then on again, and it would be OK for a short time, and then the same thing. I knew this was not a component, because if it were, it would usually not recover like that (unless there was a thermal problem which I doubted). The best procedure to follow if the above did not work, is to take a meter and check all power supply voltages at the power supply and at each regulator and be sure all voltages are present. On my H89, the schematic indicated 53 volts driving the video circuits. I checked this when my screen went dead, but the voltage was present. The other side of the flyback transformer receives a horizontal sync signal which originates on the video terminal logic board. This was not present, thus no flyback voltages were being generated (and yes, the CRT filament was also dead since it derives its power from the flyback transformer also).

I removed all interconnect cables and checked them. I usually do a good job soldering, but in my rush to get my H19 turned into an H89 I did a lousy job on a few cables I made up. I pulled on the wires, and some came out of the spring connector. I re-soldered these and all connections on the power supply PC board, sprayed all interconnect plugs with contact cleaner and re-seated them, and all was fine from then on. It is a good idea to do this to all cables to be sure they are making good contact if you are having intermittent problems of any kind.

It is also a good idea to carefully inspect all solder joints on every PC board where a problem is suspected. Don't ever think just because it's Heath built there will be no problems. The majority of problems I have had with computers were on the Heath built boards, not on my own. I have found cold solder joints, IC sockets installed with a pin bent under, and even IC's inserted with a bent pin in such a manner that it looked OK, and it worked sometimes. When the IC would become jarred slightly, the bent pin would not make contact with the socket and that board was down.

Perhaps the major problem of systems is with the disk drives. This is only expected since it is partly mechanical, operates on very tight specs, and has a 34 pin (or 50 for 8" drives) connector to worry about. These are all things which can cause problems.

I cannot stress too much to backup all disks! Keeping Murphy's law in mind, as long as you back up disks, you'll probably never need to use them. The one time you do not, you will destroy a disk and that's the end of that. Keeping backup disks for everything, and many backup disks for important things will usually prevent problems. If problems do occur, you'll at least have a backup.

When a disk drive acts up, the most obvious item to check is rotational speed. If the drives become out of tolerance (or a tight disk is loading them) they naturally won't work properly. I have found Verbatim disks to be the worse. They seem to get tight after awhile. I use Memorex for all my 5" and have had great success with them. I recommend Dysan for the 8" drives. Besides using TEST 17, I find the best way to rapidly check a drive speed is to view it under a fluorescent light. In my office that's all I have, so it's an easy matter to pop off the cover of my H17 or H89 and just look at the stroboscopic disk on the bottom of the large pulley. If the speed is 300 RPM (360 for 8") the strobe disk lines will appear to stand still, if not the lines will be in motion. Adjust the speed pot till they stand still, and your all set-- sweet and simple.

For those who insist on using floppy's, too bad, your heads are probably clogging up. Those using head cleaner disks, again, sorry. That's like cleaning your heads with sandpaper! Drive manufacturers recommend against them, and some will void warranties if they are used. Get a proper head cleaner and carefully clean them. On the Tandon type drives with no head load solenoid, don't ever close the door without a diskette inserted, as it causes the heads to slam together, and may eventually scratch or knick them. When you are finished using this type of drive, leave the door open! Disk mounting? I found in tests I have run on many different types of drives that horizontal mount drives function better than vertical mount drives, and they seem to last longer also. The only vertical drives found around here are the ones in the H89's. I found with a vertical mounted drive, more of a load is placed on the motor and I get more soft errors. There have been drives sent to me that "would not work right". I checked them for hours with not even a soft error. Then I'd tip them on their sides, and the problems would show up. It seems HDOS is better for recovering errors than CP/M, and unless you do a STAT with a comprehensive STAT program, you'd never know.

Another thing to check is step rate. As some drives begin to get old, they may not accurately step as fast as they once did. If you begin to have reliability problems, and sometimes it works good and sometimes not, and you have checked speed, then it's time to run TEST17 to see how fast it checks out, and adjust the values accordingly, or at least lubricate critical parts such as stepper screws, or head guides. The best lubricant to use is TRI-FLON which is available at most gun smith shops or large hardware shops. Use ever so little, and apply the spray to a screw driver tip and from there transfer to the part to be lubricated. NEVER use a spray in the drive itself. If you get it on the heads or cone mechanism, you may be in for real trouble.

The power supply is another item to check. Be sure you have within 10% the proper 5 and 12 Volt supplies. Also be sure on your drives to only have ONE terminating resistor pack installed per set of drives. Although this should be installed on the last physical cable location for the drives (or on the high density drive if a mixture of drive types are used), I find it's easier to keep the one installed in the H89's and remove all the rest. This has caused no problems thus far.

A problem which is reported frequently is the disk drive cable. This cable has the connector ends attached by the connector making physical contact with the wire in the cable. It is not soldered, and can cause great problems. If

your drive doesn't come up, or works sometimes and sometimes not, check your cable. Make a resistance check of each connector, pin for pin. While doing this, move the cable around and see if the resistance changes or jumps. If it does, you may have cable problems. This will be a long job, but worth it. If you have another cable, substitute it.

I have presented here some items to check which are usually overlooked. One usually wants to suspect a bad IC or other component, and usually this is not the case. Follow these guidelines, and you can probably eliminate most of your intermittent type problems.

BIOS80 4MHZ PATCHES

For those of you using the BIOS-80 CP/M mods with the Tandon drives and wanting to go 4MHz, here it is. There are two versions, one for the .02 CP/M and one for the .03 CP/M. They were both provided by the author of the BIOS80, Ray Livingston of Livingston Logic Labs. It is important to note that both of these patches are for 4Mhz only. The distribution version is for 2MHz, use one or the other. This patch once installed will not allow 2MHz operation, only 4Mhz.

For the CP/M 2.2.02 BIOS-80 version, make these patches to BIOS80.SYS. All address values are relative to the start of the file.

CONST	LOCATION	OLD	NEW
WRITA	08B3H	14H	2DH
WRITC	08BBH	10H	2DH
WHDA	0AE8H	14H	2EH
WHNA	0AF3H	14H	2EH
READA	0B08H	30H	6BH
WSCA	0B14H	50H	97H

Patch the Heath/ Zenith H-17 BLDR for 2/4 Mhz operation as follows: The location given is relative to the start of the BLDR. This patch will work at any CPU speed up to 4Mhz and need not be changed back for 2MHz operation.

WSCA	0440H	50H	97H
(23C0H) in MOVCPM image			

For the 2.2.03 CP/M BIOS mod for 4MHz operation, make these patches to BIOS80V2.SYS. All address values are relative to the start of the file.

CONST	LOCATION	OLD	NEW
WRITA	0775H	14H	2DH
WRITC	077DH	10H	2DH
WHDA	0836H	14H	2EH
	09D7H	14H	2EH
WHNA	09E2H	14H	2EH
READA	09F7H	30H	6BH
WSCA	0A03H	50H	97H

SAVE size =21

Patch to Heath/ Zenith H-17 BLDR for 2/4 MHz operation as follows: The location given is relative to the start of the BLDR. This patch will work at any CPU speed up to 4MHz and need not be changed for 2Mhz operation.

WSCA	0336H	50H	97H
(22B6H) in MOVCPM image			

Sorry, ran out of room for Classifieds

May deadline March 13